
Counting

UTSA Problem Solving Club • Fall 2026

Everybody Knows Somebody

At a gathering of $n \geq 2$ people, some pairs are friends (friendship is mutual). Show that two people at the gathering have exactly the same number of friends present.

Cutting Up the Plane

Draw n straight lines across the whole plane so that no two are parallel and no three pass through a common point. Into how many regions do they cut the plane?

Even Parity

A string of n zeros and ones has *even parity* if it contains an even number of ones. How many strings of length n have even parity?

Find the answer in two different ways.

Count the Opposite

(a) How many odd four-digit numbers contain at least one even digit?

(b) How many whole numbers below 1,000,000 have a 2 somewhere in their decimal expansion?

Routes Across Town

A city's streets form a grid. You walk from the southwest corner of a 6×6 block of streets to the northeast corner, always moving east or north along the streets.

(a) How many different routes are there?

(b) How many of them avoid the central intersection?

Forty-Three Visits

A social worker makes 43 visits over the course of 22 days, making at least one visit every day. Must there be a run of consecutive days during which she makes exactly 21 visits?

Same question, but over the course of 23 days.

OPTIMIZATION

How many different strings of letters can be made by rearranging all the letters of

OPTIMIZATION?

Around a Round Table

(a) In how many ways can n people sit around a round table, if two seatings count as the same when one is a rotation of the other?

(b) What if the table may also be turned over, so that a seating and its mirror image count as the same?

Going Further

Nobody Waits for Change

A cinema charges \$5 a ticket, and the till starts completely empty. Ten people queue up: five carry a single \$5 note, and five carry a single \$10 note. A person with a \$10 note can only be served if the till already holds a \$5 note to hand back.

- (a) In how many of the possible orders of the queue is everyone served without anyone waiting for change?
 (b) Answer the same question for n people of each kind.

Thirty-Nine on an Icosahedron

Recall that a regular icosahedron is a convex polyhedron having 12 vertices and 20 faces; the faces are congruent equilateral triangles. On each face of a regular icosahedron is written a nonnegative integer such that the sum of all 20 integers is 39. Show that there are two faces that share a vertex and have the same integer written on them.

Source: Putnam 2013 A1

Sliding Coins

Consider an m -by- n grid of unit squares, indexed by (i, j) with $1 \leq i \leq m$ and $1 \leq j \leq n$. There are $(m-1)(n-1)$ coins, which are initially placed in the squares (i, j) with $1 \leq i \leq m-1$ and $1 \leq j \leq n-1$. If a coin occupies the square (i, j) with $i \leq m-1$ and $j \leq n-1$ and the squares $(i+1, j)$, $(i, j+1)$, and $(i+1, j+1)$ are unoccupied, then a legal move is to slide the coin from (i, j) to $(i+1, j+1)$. How many distinct configurations of coins can be reached starting from the initial configuration by a (possibly empty) sequence of legal moves?

Source: Putnam 2023 B1

The Mad Teacups

The *Mad Tea Party* is a theme-park ride: a large disk carrying 3 turntables, each carrying 6 teacups, each seating 4 people — so $3 \times 6 \times 4 = 72$ riders at once. The cups spin, the turntables spin, and the disk spins, all at once. Two seatings count as the same if one becomes the other by rotating people within a cup, rotating the cups on a turntable, or rotating the turntables on the disk.

- (a) Count the seatings of one cup: 4 people, which may be rotated.
 (b) Count the seatings of one turntable: 6 cups of 4, so 24 people, where cups may be rotated *and* the turntable may be rotated.
 (c) Count the seatings of the whole ride, all 72 riders.
 (d) Now m turntables, each with n teacups, each seating k people. Find the number $a_{m,n,k}$ of seatings of the full ride of mnk people.

